

Technical Document #540: Data Engineering - Comprehensive Overview

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ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

Stream processing handles continuous data streams in real-time. Apache Kafka and Apache Flink are popular stream processing frameworks.

Schema evolution handles changes in data structure over time. Schema registries manage schema versions and ensure backward compatibility.

Data warehouses store structured data optimized for analytical queries. They use columnar storage and support complex aggregations.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Key Takeaways:

- Data lineage tracks the origin and movement of data through systems.
- Data quality ensures data is accurate, complete, and consistent.
- ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.